

**This assignment will not be graded and is only for practice.**

### Level 1

1) Consider two vectors  $a = (2, 0, 3, 0, 8)$ ,  $b = (3, 2, -2, 4, 0)$  in  $\mathbb{R}^5$ . Choose the set of correct options. **1 point**

- ☐  $a$  and  $b$  are orthogonal.
- ☐  $a$  and  $b$  are not orthogonal.
- ☐  $(a - b) \cdot a = 0$ .
- ☐  $(a - b) \cdot a = 77$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$a$  and  $b$  are orthogonal.

$(a - b) \cdot a = 77$ .

2) Choose the set of correct statements.

**1 point**

- ☐ In an orthogonal set, the norms of all the vectors are equal.
- ☐ In an orthogonal set, the vectors are linearly independent.
- ☐ In an orthogonal set, the vectors are linearly dependent.

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $\mathbb{R}^n$ , then there must be a unique solution to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $\mathbb{R}^n$ , then there are no solutions to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

- ☐ The determinant of a square matrix formed by a set of orthogonal vectors in  $\mathbb{R}^n$  is zero.
- ☐ A set of  $n$  vectors can never form an orthogonal basis in  $\mathbb{R}^{n-1}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

In an orthogonal set, the vectors are linearly independent.

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $\mathbb{R}^n$ , then there must be a unique solution to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

A set of  $n$  vectors can never form an orthogonal basis in  $\mathbb{R}^{n-1}$ .

3) Which of the following is an orthogonal basis of the given vector spaces with respect to the standard inner product (dot product)? **1 point**

- ☐  $\{(1, 0), (0, 1)\}$  is an orthogonal basis of  $\mathbb{R}^2$ .
- ☐  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$  is an orthogonal basis of  $\mathbb{R}^3$ .
- ☐  $\{(3, 4), (4, -3), (2, -3)\}$  is an orthogonal basis of  $\mathbb{R}^2$ .
- ☐  $\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$  is an orthogonal basis of  $\mathbb{R}^3$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0), (0, 1)\}$  is an orthogonal basis of  $\mathbb{R}^2$ .

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$  is an orthogonal basis of  $\mathbb{R}^3$ .

## Level 2

4) Find a vector in  $\mathbb{R}^4$  that is orthogonal to the subspace spanned by  $(1, 1, 0, 0)$  and  $(0, 1, 1, 0)$  **1 point** with respect to the dot product as the inner product.

- ☐  $(1, -1, 1, 0)$
- ☐  $(2, 3, 4, 5)$
- ☐  $(1, -1, 1, 1)$

☐ (1,1,1,0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,-1,1,0)

(1,-1,1,1)

5) Which of the following are orthogonal to the vector  $(1, 2)$  in  $\mathbb{R}^2$  with respect to the inner product  $\langle v, w \rangle = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$ ?

**1 point**

☐  $(-7, 1)$

☐  $(6, -1)$

☐  $(7, 1)$

☐  $(-9, 1)$

☐  $(14, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, 1)$

$(14, 2)$

6) Consider the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9.$$

Number of solutions of the above system is

No, the answer is incorrect.

Score: 0

## Accepted Answers:

(Type: Numeric) 1

**1 point**7) Let  $A$  be the coefficient matrix of the system of linear equations:**1 point**

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9.$$

Which one of the following is true about the matrix  $AA^T$ ?

- ☐ A scalar matrix.
- ☐ The identity matrix.
- ☐ A diagonal matrix.
- ☐ A lower triangular matrix.
- ☐ An upper triangular matrix.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

A diagonal matrix.

A lower triangular matrix.

An upper triangular matrix.

Your score is: 0/7